

TANISHA GUPTA

AI Engineer | GenAI & Machine Learning

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Professional Summary

AI/ML undergraduate specializing in Generative AI, LLMs, RAG, and Computer Vision. Research Intern at IIT (BHU), reproducing CVPR 2026 vision-language tracking research, with experience leading Generative AI and full-stack initiatives at The Wall of Dreams. Built AI systems spanning agentic LLM workflows, RAG & multimodal intelligence using Python, PyTorch, FastAPI & React.

Technical Skills

Languages	Python, JavaScript, TypeScript, C
Generative AI / LLMs	LLMs (GPT, Mistral, LLaMA, Qwen-VL), Prompt Engineering, RAG, Vector Embeddings, Agentic AI, Vision-Language Models
Machine Learning	Machine Learning, Deep Learning, Computer Vision, Transformers, PyTorch, NumPy, Pandas, Scikit-learn
Frameworks & APIs	LangChain, Hugging Face, OpenAI API, Cohere, FastAPI, Flask, React, SQLAlchemy
Web & Data	HTML, CSS, Web Scraping & Data Collection (BeautifulSoup)
Databases & Infra	PostgreSQL, MySQL, ChromaDB, Redis, Git/GitHub, Linux, AWS, Render, Vercel

Education

B.Tech, Computer Science Engineering (AI & ML)

Ashoka Institute of Technology and Management, Varanasi

Oct 2023 – June 2027

SGPA: 8.71 / 10

Professional Experience

Team Lead — Generative AI & Full-Stack Development, The Wall of Dreams

Jan 2026 – Mar 2026

- Led a cross-functional development team as Team Lead on client-facing engagements, architecting and deploying full-stack solutions that bridged Generative AI integrations with scalable web applications.
- Directed sprint planning and technical execution across project milestones, mentoring peer developers and coordinating requirements to ensure on-time delivery; technical ownership and leadership formally recognized in a client **Letter of Recommendation**.

Research Experience

Research Intern — IIT (BHU), Varanasi

May 2026 – Present

RAGTrack (CVPR 2026) — Language-Aware RGB-Thermal Object Tracking

- Reviewed 6+ SOTA Vision-Language and RGB-Thermal tracking papers, identifying research gaps in multimodal fusion, temporal reasoning, and language-guided target adaptation.
- Reproduced the RAGTrack architecture in PyTorch, integrating a Multi-modal Transformer Encoder, Adaptive Token Fusion, and Context-aware Reasoning Module with Retrieval-Augmented Generation, CLIP, and Qwen2.5-VL.
- Engineered training/inference/evaluation pipelines for the LasHeR benchmark (1,224 RGB-T video pairs, 730K+ frame pairs) and RGBT210 using Vision Transformers and multi-GPU Linux workflows; validated the reproduced tracker on 210 sequences / 104,706 frames, achieving **86.16% Precision (PR@20px)** and **58.89% Success (AUC)**.
- Proposed a Behavioral Reasoning & Adaptive Fusion enhancement to strengthen cross-modal interaction and improve robustness to occlusion, illumination variation, and modality-specific noise.

Projects

CodeMentor AI — AI-Native Software Engineering Platform

[GitHub](#) | [Live Demo](#)

FastAPI, React/TypeScript, PostgreSQL, Redis, ChromaDB, LLMs, RAG

- Architected a full-stack platform from scratch (20+ backend modules, 25+ frontend pages) integrating 6+ AI-powered workflows for coding practice, AI code review, debugging, and personalized mentoring.
- Built self-populating AI content and RAG pipelines with ChromaDB and Cohere embeddings, enabling on-demand generation and persistent retrieval of grounded, source-cited mentoring knowledge.
- Developed a custom Python execution tracer (**sys.settrace**) and an asynchronous LLM-powered code-review pipeline via FastAPI background tasks for non-blocking, line-by-line execution visualization.
- Hardened authentication with JWT access/refresh token rotation, bcrypt hashing, and Redis-based rate limiting; deployed end-to-end on Render and Vercel.

AI Video Intelligence Platform — Multimodal RAG & Video Understanding

[GitHub](#)

Python, Whisper, Mistral AI, LangChain, ChromaDB, Sentence Transformers, Streamlit

- Engineered an end-to-end multimedia intelligence pipeline supporting 4+ input sources and 10+ audio/video formats, converting YouTube videos, podcasts, and meetings into searchable knowledge using Whisper and RAG.
- Built a retrieval-grounded knowledge layer with Sentence Transformers and ChromaDB for semantic search and conversational QA over transcribed multimedia content.
- Developed LLM-powered analysis generating 8 structured outputs (summaries, key insights, meeting minutes, action items) using Mistral AI, with automated media processing via yt-dlp and FFmpeg.

Achievements

- Ranked in the Top 100 (92nd) in the **UnvibeCode Engineering Challenge 2026** (Alphashots.ai), an open-source engineering challenge evaluating complex codebases through business workflows, connected code, and evidence-backed risk analysis.
- Secured **Global Rank 7907** in **TCS CodeVita Season 13** (cleared Round 1), a national-level competitive programming contest, demonstrating strong coding and problem-solving skills.
- 200+ DSA problems solved across LeetCode, CodeChef, and Blind 75.

Certifications

Data Analytics Simulation — Tata Group via Forage (2026) | Claude 101 — Anthropic (2026) | AI & Data Science — Infosys Springboard | Microsoft Power Apps & Dataverse (2025)